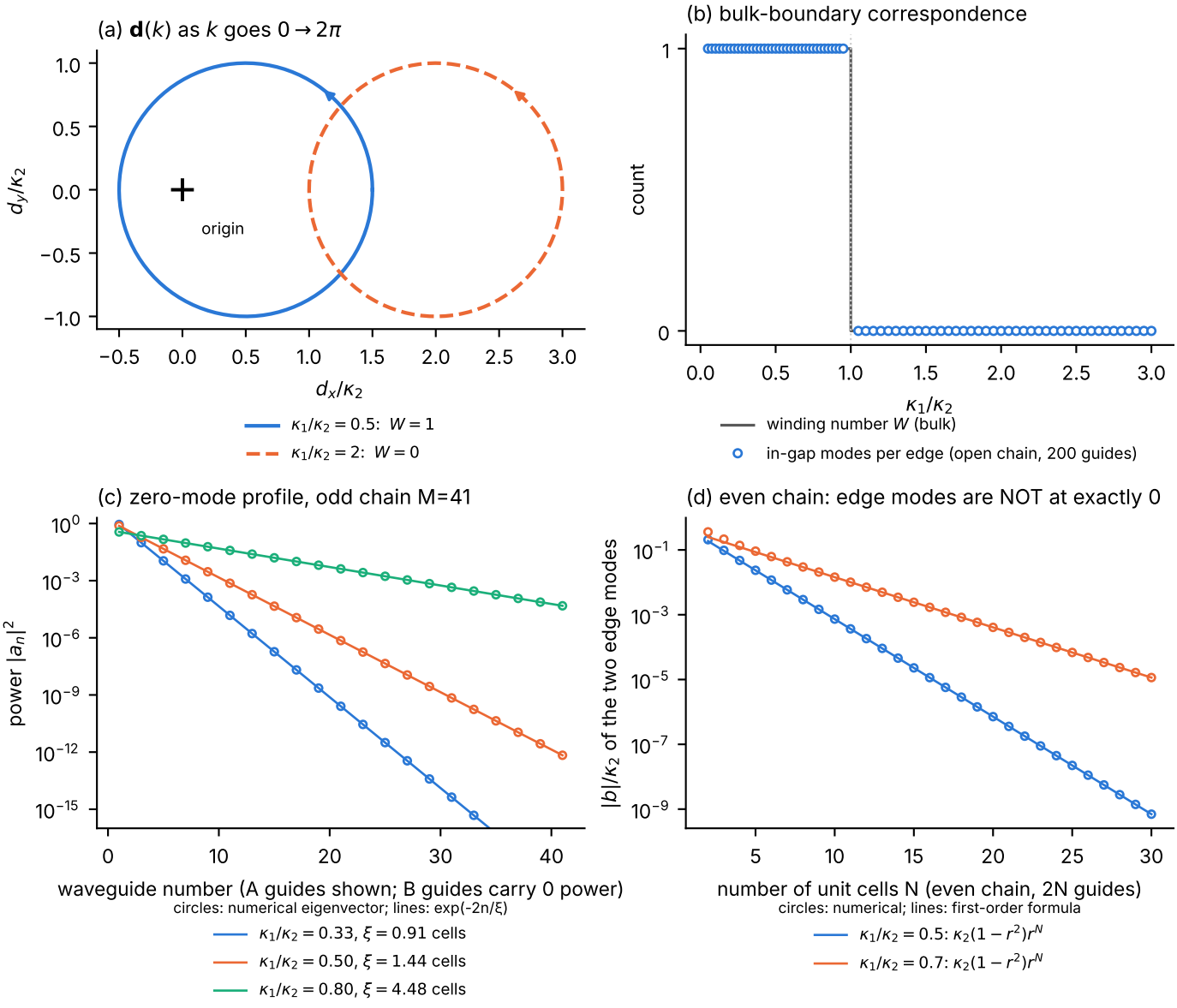
(a) d(k) as k goes 0 → 2π
origin
d_y/κ_2
d_x/κ_2
$\kappa_1/\kappa_2 = 0.5$: $W = 1$
$\kappa_1/\kappa_2 = 2$: $W = 0$
(b) bulk-boundary correspondence
count
κ_1/κ_2
winding number W (bulk)
in-gap modes per edge (open chain, 200 guides)
(c) zero-mode profile, odd chain M=41
power $|a_n|^2$
waveguide number (A guides shown; B guides carry 0 power)
circles: numerical eigenvector; lines: exp(-2n/ξ)
$\kappa_1/\kappa_2 = 0.33$, $\xi = 0.91$ cells
$\kappa_1/\kappa_2 = 0.50$, $\xi = 1.44$ cells
$\kappa_1/\kappa_2 = 0.80$, $\xi = 4.48$ cells
(d) even chain: edge modes are NOT at exactly 0
$|b|/\kappa_2$ of the two edge modes
number of unit cells N (even chain, 2N guides)
circles: numerical; lines: first-order formula
$\kappa_1/\kappa_2 = 0.5$: $\kappa_2(1-r^2)r^N$
$\kappa_1/\kappa_2 = 0.7$: $\kappa_2(1-r^2)r^N$